

5 The Structure and Function of Large Biological Molecules

KEY CONCEPTS

- 5.1** Macromolecules are polymers, built from monomers *p. 67*
- 5.2** Carbohydrates serve as fuel and building material *p. 68*
- 5.3** Lipids are a diverse group of hydrophobic molecules *p. 72*
- 5.4** Proteins include a diversity of structures, resulting in a wide range of functions *p. 75*
- 5.5** Nucleic acids store, transmit, and help express hereditary information *p. 84*
- 5.6** Genomics and proteomics have transformed biological inquiry and applications *p. 86*

Study Tip

Make a visual study guide: For each class of biological molecules, draw two examples and list their structural similarities and their functions.

Important Biological Molecules	
Carbohydrates	Proteins
Nucleic acids	Lipids

Go to Mastering Biology

For Students (in eText and Study Area)

- Animation: Making and Breaking Polymers
- Figure 5.11 Walkthrough: The Structure of a Phospholipid
- BioFlix® Animation: Gene Expression

For Instructors to Assign (in Item Library)

- Molecular Model: Lysozyme
- Tutorial: Nucleic Acid Structure

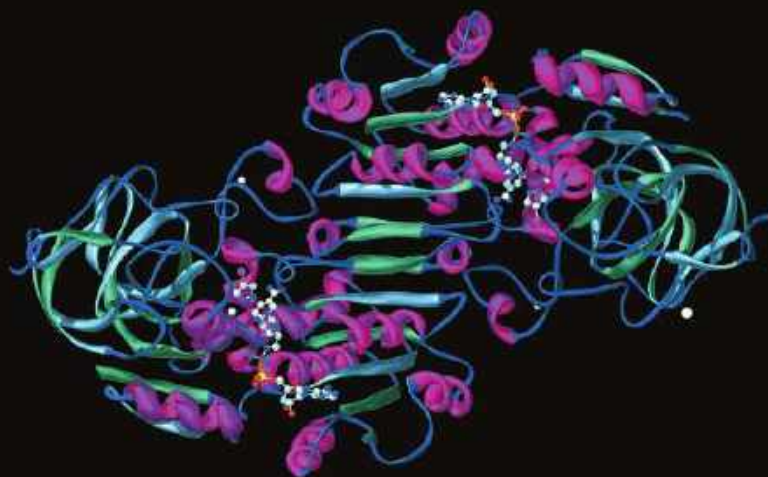
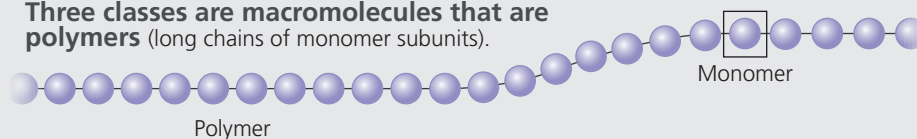


Figure 5.1 Alcohol dehydrogenase, a protein that breaks down alcohol in the body, is shown here as a molecular model. The form of this protein that an individual possesses affects how well that person tolerates drinking alcohol. Proteins are one class of large molecules, or macromolecules.

What are the structures and functions of the four important classes of biological molecules?

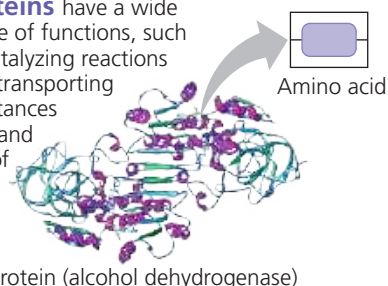
Three classes are macromolecules that are **polymers** (long chains of monomer subunits).



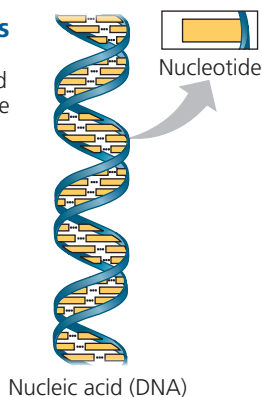
Carbohydrates are a source of energy and provide structural support.



Proteins have a wide range of functions, such as catalyzing reactions and transporting substances into and out of cells.



Nucleic acids store genetic information and function in gene expression.



The fourth class, **lipids**, are not polymers or macromolecules.

Lipids are a group of diverse molecules that do not mix well with water. Key functions include providing energy, making up cell membranes, and acting as hormones.

